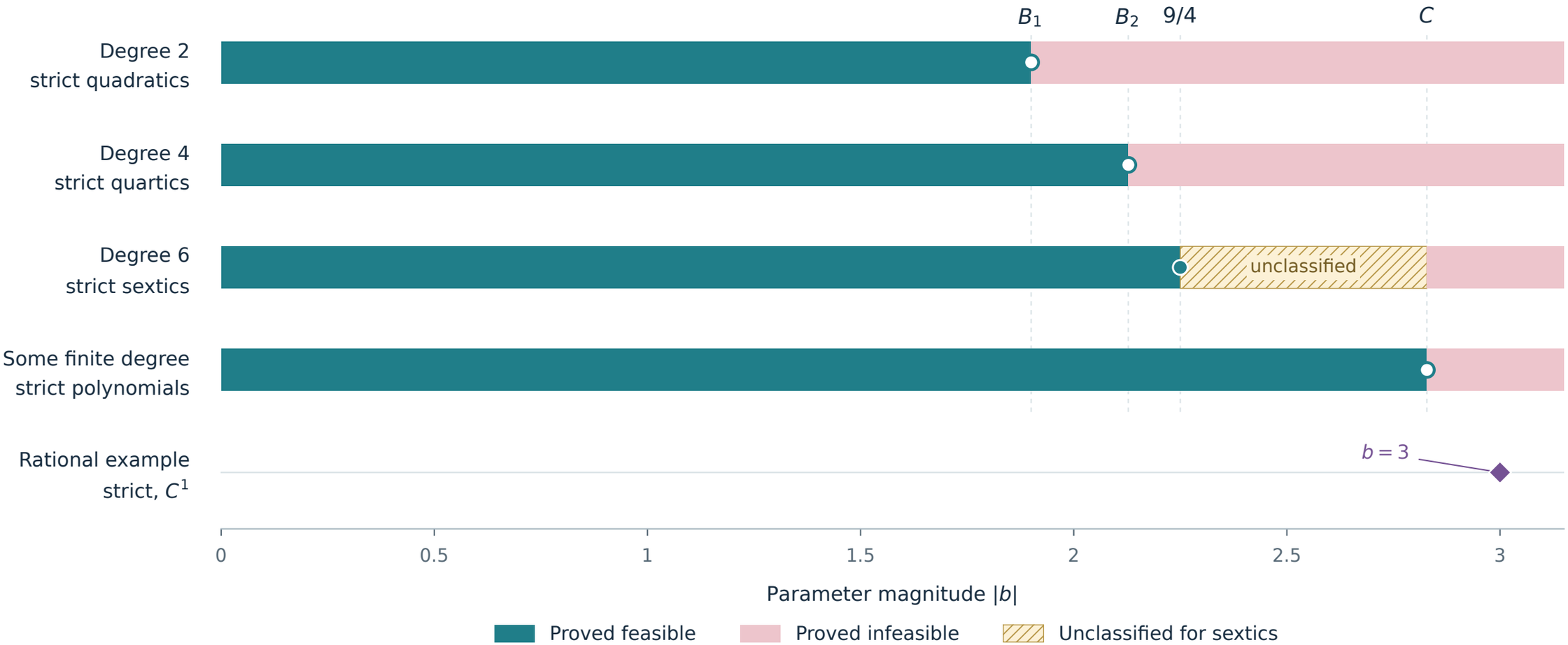
Polynomial certificate regions at damping one

Exact boundaries, a certified sextic interval, and the rational counterexample



Quadratic endpoint
$B_1 \approx 1.90018909935$

Quartic endpoint
$B_2 \approx 2.12815353309$

Certified sextic point
$9/4 = 2.25$

All-degree boundary
$C = 2\sqrt{2} \approx 2.828427$

Endpoint markers Open circles exclude strict endpoints; the filled circle includes the sextic point.

Weak certificates Quadratics include B_1 and quartics include B_2. The all-degree endpoint C is excluded even in the weak case.

Sextic limit The exact boundary B_3 is unclassified; the strict certificate at 9/4 proves $B_3 > 9/4$.

Rational example At $b = 3$, a strict C^1 rational certificate exists, while every local real-analytic weak certificate is excluded.

Quartic classification and displayed pointwise sextic certificate: Lean; other analytic regions: written.